



GLOBAL ECONOMICS
GRADE: 11TH
TERM 3 – LEARNING EVIDENCE 2
HYPOTHESIS TEST IMPLEMENTATION (C4 – C5)
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**THIRD
TERM
2025 – 2026**

Learning objective: Apply one-sample hypothesis testing procedures in context by computing test statistics, making statistical decisions, and writing contextual conclusions for both proportion and mean settings.

Evaluated criteria:

C4: Calculates hypothesis tests about a population proportion.

C5: Calculates hypothesis tests on the mean of a population.

Criteria assessment

Each assessment criterion is evaluated through the four required implementations. A criterion is considered passed when it is correctly applied in its two required implementations.

For a set of tasks to count as valid and complete, it must include:

- Exactly 2 proportion-test implementations for Criterion C4 and exactly 2 mean-test implementations for Criterion C5
- For each of Criteria C4 and C5, include one implementation in which the data and test are expected to lead to rejecting H_0 , and one implementation in which the data and test are expected to lead to not rejecting H_0 .
- At least 1 implementation based on real-world data collection, such as a survey, direct observation, counting, timing, or recording measurements. At least 1 implementation based on web databases or publicly available online datasets
- Enough evidence of the data-obtention process to verify the work, such as tables, screenshots, survey forms, observation records, database references, or clearly organized notes

Student Problems

Use the following shared instructions for all six problems. From the six contexts below, complete exactly 4 implementations in total, following all validity rules above.

- **(C4–C5)** For each implementation: describe the context; state whether it is a population proportion test or a population mean test; define the parameter and the benchmark claim; explain how the data will be obtained; present the actual sample data or summary sample information; show evidence of the data-obtention process; choose a significance level; write H_0 and H_a with correct notation; perform the hypothesis-test calculations; state the final statistical decision; and write the conclusion in context.

1. Problem 1 — School context

Create a hypothesis test about a measurable feature of your school.

2. Problem 2 — Home context

Create a hypothesis test about a measurable feature of life at home.

3. Problem 3 — Neighborhood context

Create a hypothesis test about a measurable feature of your neighborhood.

4. Problem 4 — Your Grade 11 group context

Create a hypothesis test about a measurable feature of your own Grade 11 group.

5. Problem 5 — Personal context

Create a hypothesis test about a measurable feature related to yourself.

6. Problem 6 — Bogotá context

Create a hypothesis test about a measurable feature of Bogotá that can be investigated at your level.